

 YTU – Faculty of Chemical and Metallurgical Engineering Final Exam- Questions and Answers Sheet	NOTE CHART			
	1. q	2. q	3. q	Total
Student Name and Surname				
Students Number				
Section	Department of Mathematical Engineering			Exam Date 17/12/2015
Course Name	Linear Algebra	Group Number	Exam Duration	Examination Room
Course Instructor Name and Surname	Assoc. Prof. FATMA AYDIN AKGUN			Signature
Student Disciplinary Regulations "and to make or attempt to make copies of exams to" the actual perpetrators are suspended from one or two semesters. (YÖK; 2547 Student Disciplinary Regulations, 9. Article)				

Question 1 (25 points)

Solve the following linear equation system using Cramer Method.

$$x + 2y + 3z = 2$$

$$2x + y + 3z = 5$$

$$3x + 2y + z = 6$$

Question 2 (25 points)

In the set of two dimensional vectors $V = \{(x, y) : (x, y) \in \mathbb{R}^2\}$, the operations addition and multiplication with a scalar are defined respectively as following.

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

$$k(x, y) = (x, ky)$$

Is V a vector space?

Question 3 (30 points)

The matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & -3 & 0 \\ 3 & 0 & 2 \end{bmatrix}$ is given.

- Find the eigenvalues of the matrix A.
- Find the corresponding eigenvectors.
- Determine if the matrix A diagonalizable or not, if it is, then diagonalize the matrix A.

Question 4 (20 points)

- Show that the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & -3 & 0 \\ 3 & 0 & 2 \end{bmatrix}$ (given in the Question 3), satisfies the Cayley Hamilton theorem.
- If A has an inverse, find the inverse of matrix A using Cayley Hamilton Theorem.

Question 4 (20 points)

Let $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$ be two matrices in 2×2 Vector Space M. We define

$$\langle A, B \rangle = a_1b_1 + a_2b_2 + a_3b_3 + a_4b_4.$$

Show that this gives an inner product on M.

I wish you success.

① $x + 2y + 3z = 2$
 $2x + y + 3z = 5$
 $3x + 2y + z = 6$

$$\underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_X = \underbrace{\begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}}_B$$

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & -3 & -3 \\ 0 & -4 & -8 \end{vmatrix} = 1 \begin{vmatrix} -3 & -3 \\ -4 & -8 \end{vmatrix} = 12 //$$
 (5)

$$\Delta x = \begin{vmatrix} 2 & 2 & 3 \\ 5 & 1 & 3 \\ 6 & 2 & 1 \end{vmatrix} = \begin{vmatrix} -8 & 0 & -3 \\ 5 & 1 & 3 \\ -4 & 0 & -5 \end{vmatrix} = \begin{vmatrix} -8 & -3 \\ -4 & -5 \end{vmatrix} = 40 - 12 = 28$$
 (5)

$$\Delta y = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 3 & 6 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & -8 \end{vmatrix} = -8$$
 (5)

$$x = \frac{28}{12} //$$
 $y = \frac{-8}{12} //$

$$\Delta z = \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & 5 \\ 3 & 2 & 6 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 2 \\ 0 & -3 & 1 \\ 0 & -4 & 0 \end{vmatrix} = 4$$
 (5)

$$z = \frac{4}{12} = \frac{1}{3} \quad x = \frac{7}{3} \quad y = \frac{-2}{3}$$
 (5)

2) ④ $(x_1, y_1) + (x_2, y_2) \in \mathbb{R}^2$

$v_1 + v_2 = v_2 + v_1$

1) $(x_1, y_1) + (x_2, y_2) = (x_2, y_2) + (x_1, y_1) ?$

$\parallel$
 $(x_1 + x_2, y_1 + y_2) = (x_2 + x_1, y_2 + y_1) = (x_2, y_2) + (x_1, y_1)$

2) $(x_1, y_1) + [(x_2, y_2) + (x_3, y_3)] \stackrel{?}{=} [(x_1, y_1) + (x_2, y_2)] + (x_3, y_3)$

$(x_1, y_1) + (x_2 + x_3, y_2 + y_3) = (x_1 + x_2 + x_3, y_1 + y_2 + y_3)$
 $= ((x_1 + x_2) + x_3, (y_1 + y_2) + y_3)$
 $= (x_1 + x_2, y_1 + y_2) + (x_3, y_3)$
 $= [(x_1, y_1) + (x_2, y_2)] + (x_3, y_3)$

3) $0 \in \mathbb{R}^2$ ^{now} $0 + (x, y) = (x, y)$ olur.

$(0, 0) \in \mathbb{R}^2$ ✓

4) $v + -v = 0$ olacak şekilde $v \in \mathbb{R}^2$ varmı?

$(-x, -y) + (x, y) = (0, 0)$

$-x, -y \in \mathbb{R} \rightarrow (-x, -y) \in \mathbb{R}^2$ ✓

(10 puan) • $k(x, y) \in \mathbb{R}^2$ olmalı
 $k(x, y) = (x, ky)$ $x \in \mathbb{R}, ky \in \mathbb{R} \rightarrow (x, ky) \in \mathbb{R}^2$

1) $k(v_1 + v_2) = kv_1 + kv_2$?

$k((x_1, y_1) + (x_2, y_2)) = k(x_1 + x_2, y_1 + y_2) = (x_1 + x_2, ky_1 + ky_2)$

$kv_1 + kv_2 = k(x_1, y_1) + k(x_2, y_2) = (x_1, ky_1) + (x_2, ky_2)$ ✓
 $= (x_1 + x_2, ky_1 + ky_2)$

Toplam
30 puan

2) $(k_1 + k_2)v = k_1v + k_2v$

$(k_1 + k_2)(x, y) = (x, (k_1 + k_2)y) = (x, \underbrace{k_1y + k_2y}_{\neq})$

Vektör uzayı değil.

$k_1v = (x, k_1y)$ $k_1v + k_2v = (2x, k_1y + k_2y)$

$k_2v = (x, k_2y)$

20
 puan

Question 4

$$\langle A, B \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3 + a_4 b_4$$

$$(1) \langle A, A \rangle \geq 0 \quad \langle A, A \rangle = 0 \iff A = 0.$$

$$\langle A, A \rangle = a_1^2 + a_2^2 + a_3^2 + a_4^2 \geq 0.$$

$$\text{if } \langle A, A \rangle = 0 \quad a_1 = a_2 = a_3 = a_4 = 0 \Rightarrow A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0_{2 \times 2} //$$

$$(2) \langle A, B \rangle = \langle B, A \rangle$$

$$\langle A, B \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3 + a_4 b_4 = b_1 a_1 + b_2 a_2 + b_3 a_3 + b_4 a_4 = \langle B, A \rangle$$

$$(3) k \in \mathbb{R}$$

$$k \langle A, B \rangle = \langle kA, B \rangle$$

$$k \langle A, B \rangle = k a_1 b_1 + k a_2 b_2 + k a_3 b_3 + k a_4 b_4 //$$

$$kA = \begin{bmatrix} k a_1 & k a_2 \\ k a_3 & k a_4 \end{bmatrix} \quad \langle kA, B \rangle = (k a_1 b_1 + k a_2 b_2 + k a_3 b_3 + k a_4 b_4)$$

$$(4) \langle A+B, C \rangle = \langle A, C \rangle + \langle B, C \rangle$$

$$A+B = \begin{bmatrix} a_1+b_1 & a_2+b_2 \\ a_3+b_3 & a_4+b_4 \end{bmatrix} \quad \langle A+B, C \rangle = (a_1+b_1)c_1 + (a_2+b_2)c_2 + (a_3+b_3)c_3 + (a_4+b_4)c_4$$

$$= \underbrace{a_1 c_1 + a_2 c_2 + a_3 c_3 + a_4 c_4}_{\langle A, C \rangle} + \underbrace{b_1 c_1 + b_2 c_2 + b_3 c_3 + b_4 c_4}_{\langle B, C \rangle}$$

ig. Lapin uzejdr.